CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

61. Identify from the following species in which d^2sp^3 hybridization is shown by central atom :

(1) BrF_5 (2) SF_6

(3) $[Pt(Cl_4)]^{2-}$ (4) $[Co(NH_3)_6]^{3+}$

Answer (4)

Sol. Co^{3+} : $3d^6\,4s^0$

Pairing will take place and hybridization will be d^2sp^3.

62. Which structure of protein remains intact after coagulation of egg white on boiling?

(1) Tertiary (2) Quaternary

(3) Secondary (4) Primary

Answer (4)

Sol. Primary structure of protein remains intact (it represents the sequence of amino acids).

63. Choose the correct option having all the elements with d^{10} electronic configuration from the following :

(1) ^{27}Co, ^{28}Ni, ^{26}Fe, ^{24}Cr

(2) ^{46}Pd, ^{28}Ni, ^{26}Fe, ^{24}Cr

(3) ^{29}Cu, ^{30}Zn, ^{48}Cd, ^{47}Ag

(4) ^{28}Ni, ^{24}Cr, ^{26}Fe, ^{29}Cu

Answer (3)

Sol. Cu : $4s^13d^{10}$

Zn : $4s^23d^{10}$

Cd : $5s^24d^{10}$

Ag : $5s^14d^{10}$

64. The final product A, formed in the following reaction sequence is

$$Ph-CH=CH_2 \xrightarrow[\text{(iii) HBr}]{\text{(i) } BH_3,\ \text{(ii) } H_2O_2,\ \ominus OH} \xrightarrow{\text{(iv) Mg, ether, then } HCHO/H_3O^+} A$$

(1) $Ph-CH_2-CH_2-CH_3$

(2) $Ph-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3$

(3) $Ph-CH_2-CH_2-CH_2-OH$

(4) $Ph-\underset{\displaystyle CH_2OH}{\underset{|}{CH}}-CH_3$

Answer (3)

Sol.

$$Ph-CH=CH_2 \xrightarrow[(ii)\ H_2O_2/OH^-]{(i)\ BH_3} Ph-CH_2-\underset{\displaystyle OH}{\underset{|}{CH_2}}$$

$$Ph-CH_2-\underset{\displaystyle OH}{\underset{|}{CH_2}} \xrightarrow{HBr} Ph-CH_2-CH_2-Br$$

$$Ph-CH_2-CH_2-Br \xrightarrow[\text{then } H-\overset{\displaystyle O}{\overset{\|}{C}}-H\ /\ H_3O^+]{Mg,\ ether} Ph-CH_2-\underset{\displaystyle OH}{\underset{|}{CH_2}}-CH_2$$

(A)

Option (3) is correct.



65. Given below are two statements :

Statement (I) : Oxygen being the first member of group 16 exhibits only –2 oxidation state.

Statement (II) : Down the group 16 stability of +4 oxidation state decreases and +6 oxidation state increases.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **Statement I** is incorrect but **Statement II** is correct

(2) Both **Statement I** and **Statement II** are incorrect

(3) **Statement I** is correct but **Statement II** is incorrect

(4) Both **Statement I** and **Statement II** are correct

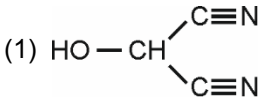
Answer (2)

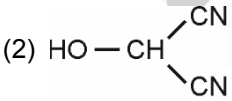
Sol. Oxygen can show other oxidation states apart from (–2).

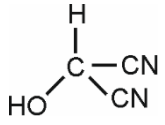
Stability of +6 oxidation state decreases down the group due to inert pair effect.

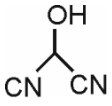
Both Statement I and Statement II are incorrect.

66. Bond line formula of $HOCH(CN)_2$ is:

(1) 

(2) 

(3) 

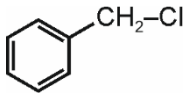
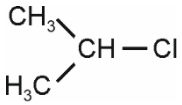
(4) 

Answer (4)

Sol. Bond line representation does not include representation of H-atoms.

67. Which among the following halide/s will not show S_N1 reaction?

(A) $H_2C=CH-CH_2Cl$ (B) $CH_3-CH=CH-Cl$

(C)  (D) 

Choose the **most appropriate** answer from the options given below:

(1) (B) and (C) only (2) (B) only

(3) (A) and (B) only (4) (A), (B) and (D) only

Answer (2)

Sol. Vinyl halide will not show S_N1 reaction, due to very less stability of carbocation formed and partial double bond character in C — Cl bond.

68. Identify the **incorrect** pair from the following:

(1) Polythene preparation - $TiCl_4$, $Al(CH_3)_3$

(2) Haber process - Iron

(3) Wacker process - $PtCl_2$

(4) Photography - AgBr

Answer (3)

Sol. Wacker process - $PdCl_2$, $CuCl_2$

69. The molecular formula of second homologue in the homologous series of mono carboxylic acids is _____.

(1) CH_2O (2) $C_2H_4O_2$

(3) $C_3H_6O_2$ (4) $C_2H_2O_2$

Answer (2)

Sol. Second homologue : CH_3COOH

$= C_2H_4O_2$



70. Identify B formed in the reaction

$$Cl-(CH_2)_4-Cl \xrightarrow{\text{excess } NH_3} A \xrightarrow{NaOH} B + H_2O + NaCl$$

(1) 

(2) $\overset{-}{Cl}\overset{+}{N}H_3-(CH_2)_4-\overset{+}{N}H_3Cl^-$

(3) 

(4) $H_2N-(CH_2)_4-NH_2$

Answer (4)

Sol. $A: Cl^-NH_3^+-(CH_2)_4-NH_3^+Cl^-$

$B: H_2N-(CH_2)_4-NH_2$

71. The incorrect statement regarding conformations of ethane is:

(1) Ethane has infinite number of conformations

(2) Eclipsed conformation is the most stable conformation

(3) The conformations of ethane are inter-convertible to one-another

(4) The dihedral angle in staggered conformation is 60°

Answer (2)

Sol. Staggered conformation is most stable

$\Rightarrow$ Option (2) is incorrect

72. The technique used for purification of steam volatile water immiscible substances is:

(1) Steam distillation

(2) Distillation

(3) Fractional distillation under reduced pressure

(4) Fractional distillation

Answer (1)

Sol. Steam distillation is used for purification of steam volatile water immiscible substances.

73. The quantity which changes with temperature is:

(1) Molarity

(2) Molality

(3) Mass percentage

(4) Mole fraction

Answer (1)

Sol. Formula of molarity includes volume which depends upon temperature.

74. Which of the following cannot function as an oxidising agent?

(1) N^{3-}

(2) SO_4^{2-}

(3) MnO_4^-

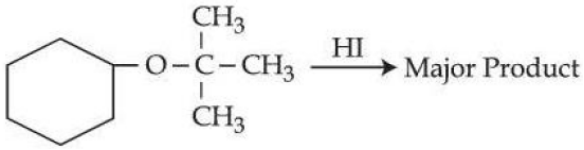
(4) BrO_3^-

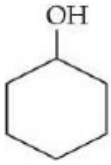
Answer (1)

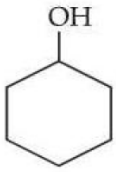
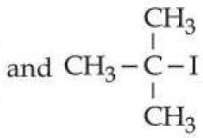
Sol. Nitrogen is present in its lowest oxidation state in N^{3-}.

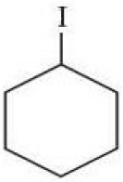


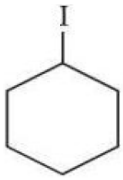
75. Major product formed in the following reaction is a mixture of:



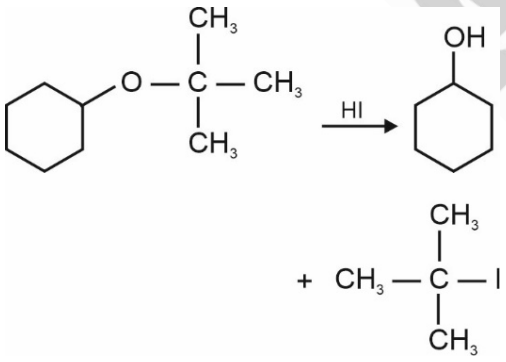
(1)  and $(CH_3)_3COH$

(2)  and 

(3)  and $(CH_3)_3COH$

(4)  and $(CH_3)_3CI$

Answer (2)

Sol. 

76. Phenolic group can be identified by a positive:

(1) Lucas test

(2) Carbylamine test

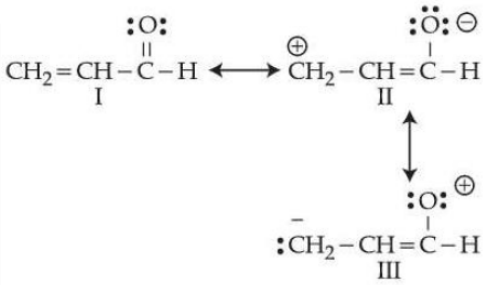
(3) Tollen's test

(4) Phthalein dye test

Answer (4)

Sol. Phenolic group is identified by phthalein dye test.

77. The order of relative stability of the contributing structure is :



Choose the **correct** answer from the options given below:

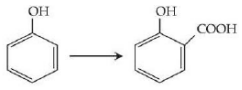
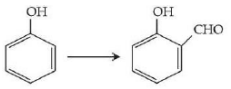
(1) III > II > I (2) I > II > III

(3) II > I > III (4) I = II = III

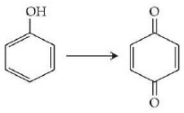
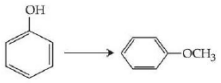
Answer (2)

Sol. In structure I, octet of every element is complete.

78. Match **List-I** with **List-II**.

	List-I (Reaction)		**List-II (Reagent(s))**
(A)		(I)	$Na_2Cr_2O_7$, H_2SO_4
(B)		(II)	(i) NaOH (ii) CH_3Cl



(C)		(III)	(i) NaOH, $CHCl_3$ (ii) NaOH (iii) HCl
(D)		(IV)	(i) NaOH (ii) CO_2 (iii) HCl

Choose the **correct** answer from the options given below:

(1) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

(2) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(3) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(4) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

Answer (2)

Sol. (A) Kolbe reaction in presence of NaOH; CO_2 and HCl.

(B) Reimer-Tiemann reaction

(C) Oxidation of phenol

(D) Williamson ether synthesis

79. Given below are two statements:

Statement (I) : In the Lanthanoids, the formation Ce^{+4} is favoured by its noble gas configuration.

Statement (II) : Ce^{+4} is a strong oxidant reverting to the common +3 state.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) **Statement I** is true but **Statement II** is false

(2) **Statement I** is false but **Statement II** is true

(3) Both **Statement I** and **Statement II** are false

(4) Both **Statement I** and **Statement II** are true

Answer (4)

Sol. $Ce_{58} = [Xe]\ 6s^2 5d^1 4f^1$

$Ce^{4+} = [Xe]\ 6s^0 5d^0 4f^0$

$E^{o}_{Ce^{4+}/Ce^{3+}}$ is very high

$E^{o}_{Ce^{4+}/Ce^{3+}} = 1.74$ V

Both **Statement I** and **Statement II** are true

80. Which of the following statements is not correct about rusting of iron?

(1) Dissolved acidic oxides SO_2, NO_2 in water act as catalyst in the process of rusting.

(2) Rusting of iron is envisaged as setting up of electrochemical cell on the surface of iron object.

(3) When pH lies above 9 or 10, rusting of iron does not take place.

(4) Coating of iron surface by tin prevents rusting, even if the tin coating is peeling off.

Answer (4)

Sol. If Tin coating is peeling off, then rusting of iron will take place as contact of iron with air will take place.

SECTION - B

Numerical Value Type Questions: This section contains 10 Numerical based questions. Attempt any 5 questions out of 10. The answer to each question should be rounded-off to the nearest integer.

81. Time required for completion of 99.9% of a First order reaction is _____ times of half life ($t_{1/2}$) of the reaction.

Answer (10)



Sol. $t_{99.9} = \frac{2.303}{K}\log\left(\frac{100}{0.1}\right)$

$= \frac{2.303}{K}\log 1000$

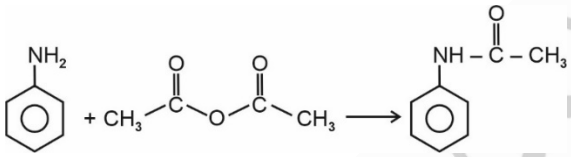
$t_{50} = \frac{2.303}{K}\log 2$

$\frac{t_{99.9}}{t_{50}} = \frac{3}{0.3} = 10$

82. 9.3 g of aniline is subjected to reaction with excess of acetic anhydride to prepare acetanilide. The mass of acetanilide produced if the reaction is 100% completed is _____ × 10^{-1} g.

(Given molar mass in g mol^{-1} N : 14, O : 16, C : 12, H : 1)

Answer (135)

Sol. 

Moles of aniline = $\frac{9.3}{93} = 0.1$

Mass of acetanilide = (0.1) × 135

= 13.5

= 135 × 10^{-1} gm

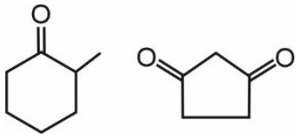
83. Volume of 3 M NaOH (formula weight 40 g mol^{-1}) which can be prepared from 84 g of NaOH is _____ × 10^{-1} dm^3.

Answer (7)

Sol. Moles of NaOH = $\frac{84}{40} = 2.1$

Volume = $\frac{\text{Moles}}{\text{Molarity}} = \frac{2.1}{3} = 0.7 = 7 \times 10^{-1}$

84. Total number of compounds with Chiral carbon atoms from following is _____.

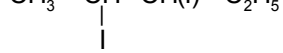


$CH_3 - CH_2 - CH(NO_2) - COOH$

$CH_3 - CH_2 - CHBr - CH_2 - CH_3$

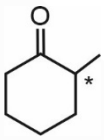
$CH_3 - CH(I) - CH_2 - NO_2$

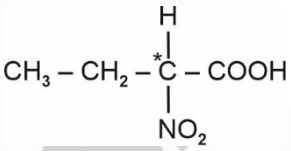
$CH_3 - CH_2 - CH(OH) - CH_2OH$

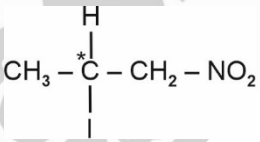


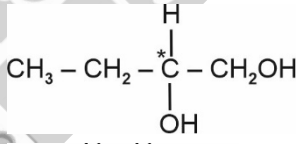
Answer (5)

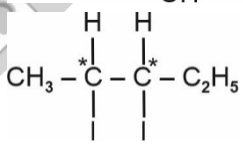
Sol. Compounds with chiral carbon are











85. The number of non-polar molecules from the following is _____.

HF, H_2O, SO_2, H_2, CO_2, CH_4, NH_3, HCl, $CHCl_3$, BF_3

Answer (4)

Sol. Non-polar molecules are

H_2; CO_2; CH_4; BF_3



86. For a certain thermochemical reaction M → N at T = 400 K, ΔH° = 77.2 kJ mol^{-1}, ΔS = 122 JK^{-1}, log equilibrium constant (logK) is – _____ × 10^{-1}.

Answer (37)

Sol. $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$= (77.2 \times 1000) - 400(122)$

$= 77200 - 48800$

$= 28400$

$\Delta G^\circ = -(2.303)(8.314)(400)\log K$

$\log K = -3.708$

$= -37.08 \times 10^{-1}$

$= 37$

87. The hydrogen electrode is dipped in a solution of pH = 3 at 25°C. The potential of the electrode will be – _____ × 10^{-2} V.

$\left(\frac{2.303\,RT}{F} = 0.059\ V\right)$

Answer (18)

Sol. $2H^+ + 2e^- \longrightarrow H_2$

$E_{cell} = 0 + \frac{.059}{2}\log(H^+)^2$

$= +.059(\log H^+)$

$= -0.059 \times PH$

$= -0.059 \times 3$

$= -0.177$ V

$= -17.7 \times 10^{-2}$ V

Nearest integer = 18

88. Total number of ions from the following with noble gas configuration is _____.

Sr^{2+} (z = 38), Cs^+ (z = 55), La^{2+} (z = 57), Pb^{2+} (z = 82), Yb^{2+} (z = 70) and Fe^{2+} (z = 26)

Answer (3)

Sol. Species with noble gas configuration;

Sr^{2+}; Cs^+; Yb^{2+}

89. The Spin only magnetic moment value of square planar complex $[Pt(NH_3)_2Cl(NH_2CH_3)]Cl$ is _____ B.M. (Nearest integer)

(Given atomic number for Pt = 78)

Answer (0)

Sol. Pt^{2+} : $5d^8\,6s^0$

Pairing will take place and hence number of unpaired electrons = 0.

90. 1 mole of PbS is oxidised by "X" moles of O_3 to get "Y" moles of O_2. X + Y = _____.

Answer (8)

Sol. $PbS + 4O_3 \longrightarrow PbSO_4 + 4O_2$

X = 4

Y = 4

(X + Y) = 8

❐ ❐ ❐

